

RESEARCH INTEREST

- **Nonlinear dynamics and Chaos theory**
- **Oscillatory associative memory**

EDUCATION

- **M.Sc. Physics** IEST, Shibpur
Gold Medalist | CGPA: 8.99 / 10 2023 – 2025
- **B.Sc. Physics** SASTRA Deemed University, Thanjavur
CGPA: 8.82 / 10 2020 – 2023
- **Ph.D Applied Mechanics** Indian Institute of Technology (IIT), Madras
2026 -

RESEARCH EXPERIENCE

- **M.Sc. Thesis – Magnetically Coupled Pendulums** IEST, Shibpur
M.Sc. Physics Aug 2024 – July 2025
 - **Theoretical Model:** Proposed a novel nonlinear model consisting of two pendulums sharing a common fulcrum and coupled through magnetic interactions, distinct from conventional serial double pendulum systems, and derived the governing equations of motion.
 - **Numerical Analysis:** Performed numerical simulations to obtain time series, phase plots, Poincaré sections, Lyapunov exponent maps, time maps of first flip, and FFT spectra.
 - **Experimental Investigation:** Conducted experiments to measure flip times and constructed time map of first flip for direct comparison with numerical predictions.
 - **Model Interpretation:** Analyzed agreement and deviations between theoretical and experimental results to identify physical mechanisms governing ordered and chaotic dynamics.
 - **Application of Chaos:** Utilized the chaotic dynamics of the system to generate random number sequences and evaluated their statistical properties through autocorrelation analysis.
- **Clean Energy Research Group – Battery Material Characterisation and testing** IIISER Bhopal
Visiting research fellow Sep 2025 – Jan 2026
 - **Cathode Preparation:** Synthesized cathode materials using standard fabrication methods and performed electrode preparation and cell assembly under argon-filled glove box conditions.
 - **Electrochemical Characterization:** Performed comprehensive electrochemical testing including galvanostatic charge-discharge (GCD), electrochemical impedance spectroscopy (EIS), galvanostatic intermittent titration technique (GITT), and entropy coefficient measurements, and analyzed the data to extract thermodynamic, kinetic, and transport parameters governing electrochemical performance.
 - **Materials Characterization:** Extracted strain components from SEM/TEM data and conducted XRD Rietveld refinement using GSAS-II with Python-based automation.
- **Defense Laboratory, Jodhpur** Defence Research & Development Organisation (DRDO)
Junior Research Fellow (JRF) Apr 2026 – Jun 2026
 - **Literature Survey:** Conducted a comprehensive literature survey and systematic study of scintillation materials for radiation detection applications.
 - **Technical Study:** Gained technical familiarity with radiation detectors, sensing technologies, and sensor systems.

PROJECTS

- **Computational Physics Codes (Python) (Dec, 2024 – March, 2025):** Developed a GitHub repository of Python codes implementing core numerical methods for standard problems in classical, quantum, and statistical physics, with emphasis on clear algorithmic structure and physical interpretation.
Github: <https://github.com/appz7/computational-physics>

TECHNICAL SKILLS

- **Computation:** Nonlinear Dynamics and Chaos, Python, Numerical simulation, Numerical data analysis, Git, GSAS-II.

ACHIEVEMENTS AND RESPONSIBILITIES

- **Oral Presentation – ICPE 2025:** Delivered an oral presentation titled *“Toy Gymnast as a Chaotic Double Pendulum”* at the International Conference on Physics Education (ICPE 2025), based on MSc thesis work in nonlinear dynamics and chaos.
- **Provisional Patent Published:** Filed and published a provisional patent titled *“Leidenfrost-Assisted Method for Synthesizing Cathode Materials for Electrochemical Energy Storage Devices,”* proposing a novel synthesis technique for cathode materials with potential applications in energy storage technologies.
- **Volunteer – NGPE Examination, Part C, 2025:** Served as a volunteer during the NGPE physics examination conducted at IEST, assisting with examination coordination and student management.
- **AIR 2812:** In GATE 2025
- **Grade 8 - Piano:** Awarded Grade 8 Piano with Distinction by Trinity College London, demonstrating long-term commitment to structured musical training.

REFERENCES

- **Dr. Sayan Gupta:** Professor (Head of Department), AMBE, IIT, Madras
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- **Dr. S Ganga Prasath:** Assistant Professor, AMBE, IIT, Madras
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